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Thermodynamics and Statistical Physics

Spring 2026

Homework 7

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Show all essential steps in your derivations.

Problem 1: Fundamental relation (adapted from Callen)***

The entropy of a single-component simple system is given as a function of U, V, N (the entropy representation):

$$S(U, V, N) = AN \ln \left(\frac{U^{3/2} V}{N^{5/2}} \right), \quad (1)$$

where A is a positive constant.

(a) Using the thermodynamic relations

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{U,N},$$

derive the equation of state $P(V, T, N)$.

(b) Use the identity

$$\mu = -T \left(\frac{\partial S}{\partial N} \right)_{U,V}$$

to derive the chemical potential μ as a function of T and P .

(c) Obtain U as a function of S, V, N (energy representation), and redo parts (a) and (b). Make sure that you obtain the same results.

(d) Compute $C_V, C_P, \kappa_S, \kappa_T$ as a function of T, V, N . Verify that stability conditions are valid.

Solution

(a) We are given the fundamental equation in the entropy representation:

$$S(U, V, N) = AN \ln \left(\frac{U^{3/2} V}{N^{5/2}} \right) = AN \left[\frac{3}{2} \ln U + \ln V - \frac{5}{2} \ln N \right]$$

To derive the equation of state, we use the given thermodynamic relations. First, differentiating with respect to U :

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N} = AN \left(\frac{3}{2} \frac{1}{U} \right) \implies U = \frac{3}{2} ANT$$

Next, differentiating with respect to V :

$$\frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{U,N} = AN \left(\frac{1}{V} \right) \implies P = \frac{ANT}{V}$$

Thus, the equation of state is the familiar ideal-gas-like relation: $PV = ANT$.

(b) The chemical potential is given by $\mu = -T \left(\frac{\partial S}{\partial N} \right)_{U,V}$. We differentiate the entropy with respect to N :

$$\begin{aligned} \left(\frac{\partial S}{\partial N} \right)_{U,V} &= A \ln \left(\frac{U^{3/2} V}{N^{5/2}} \right) + AN \left(-\frac{5}{2} \frac{1}{N} \right) \\ &= \frac{S}{N} - \frac{5}{2} A \end{aligned}$$

From part (a), we substitute $U = \frac{3}{2}ANT$ and $V = \frac{ANT}{P}$ to express S/N purely in terms of T and P :

$$\frac{S}{N} = A \ln \left(\frac{\left(\frac{3}{2}ANT\right)^{3/2} \left(\frac{ANT}{P}\right)}{N^{5/2}} \right) = A \ln \left(\frac{\left(\frac{3}{2}A\right)^{3/2} AT^{5/2}}{P} \right)$$

Therefore, the chemical potential as a function of T and P is:

$$\mu(T, P) = -T \left[A \ln \left(\frac{\left(\frac{3}{2}A\right)^{3/2} AT^{5/2}}{P} \right) - \frac{5}{2}A \right] = -AT \ln \left(\frac{\left(\frac{3}{2}A\right)^{3/2} AT^{5/2}}{P} \right) + \frac{5}{2}AT$$

(c) To obtain the energy representation $U(S, V, N)$, we invert the fundamental equation Eq. (1):

$$\frac{S}{AN} = \ln \left(\frac{U^{3/2}V}{N^{5/2}} \right) \implies \frac{U^{3/2}V}{N^{5/2}} = e^{\frac{S}{AN}} \implies U^{3/2} = N^{5/2}V^{-1}e^{\frac{S}{AN}}$$

Raising both sides to the power of $2/3$, we obtain the energy fundamental equation:

$$U(S, V, N) = N^{5/3}V^{-2/3}e^{\frac{2S}{3AN}}$$

Now we redo parts (a) and (b) using the energy representation relations: $T = (\partial U / \partial S)_{V, N}$, $P = -(\partial U / \partial V)_{S, N}$, and $\mu = (\partial U / \partial N)_{S, V}$.

$$T = \left(\frac{\partial U}{\partial S} \right)_{V, N} = U \left(\frac{2}{3AN} \right) \implies U = \frac{3}{2}ANT$$

$$P = - \left(\frac{\partial U}{\partial V} \right)_{S, N} = -U \left(-\frac{2}{3V} \right) = \frac{2U}{3V} \implies P = \frac{2 \left(\frac{3}{2}ANT \right)}{3V} = \frac{ANT}{V}$$

Both exactly match part (a). For the chemical potential:

$$\mu = \left(\frac{\partial U}{\partial N} \right)_{S, V} = \frac{5}{3N}U + U \left(-\frac{2S}{3AN^2} \right) = \frac{U}{N} \left(\frac{5}{3} - \frac{2S}{3AN} \right)$$

Substitute $U/N = \frac{3}{2}AT$ to express μ :

$$\mu = \frac{3}{2}AT \left(\frac{5}{3} - \frac{2S}{3AN} \right) = \frac{5}{2}AT - T \frac{S}{N} = -T \left(\frac{S}{N} - \frac{5}{2}A \right)$$

This identically matches the intermediate result for μ from part (b), confirming the equivalence.

(d) Now we compute the heat capacities and compressibilities as functions of T, V, N :

- **Isochoric heat capacity (C_V):**

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{V, N} = \frac{\partial}{\partial T} \left(\frac{3}{2}ANT \right) = \frac{3}{2}AN$$

- **Isobaric heat capacity (C_P):** Using $C_P = T(\partial S / \partial T)_{P, N}$. From part (b), $S = AN \ln T^{5/2} + \dots = \frac{5}{2}AN \ln T + f(P, N)$, so:

$$C_P = T \left(\frac{\partial S}{\partial T} \right)_{P, N} = \frac{5}{2}AN$$

- **Isothermal compressibility (κ_T):**

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{T,N} = -\frac{1}{V} \left(-\frac{ANT}{P^2} \right) = \frac{ANT}{P^2 V} = \frac{1}{P} = \frac{V}{ANT}$$

- **Adiabatic compressibility (κ_S):** Using $P = \frac{2}{3}N^{5/3}V^{-5/3}e^{\frac{2S}{3AN}}$ from (c), we have $P \propto V^{-5/3}$ at constant S, N .

$$\left(\frac{\partial P}{\partial V} \right)_{S,N} = -\frac{5}{3} \frac{P}{V} \implies \kappa_S = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{S,N} = \frac{3}{5P} = \frac{3V}{5ANT}$$

Stability Verification: The conditions for thermodynamic stability are thermal stability ($C_V > 0$ and $C_P > 0$) and mechanical stability ($\kappa_T > 0$ and $\kappa_S > 0$). Assuming physical states where $A, T, V, N, P > 0$, we have: $C_V = 1.5AN > 0$, $C_P = 2.5AN > 0$, $\kappa_T = V/(ANT) > 0$, $\kappa_S = 0.6V/(ANT) > 0$. Additionally, $C_P > C_V$ and $\kappa_T > \kappa_S$ are strictly satisfied. Hence, all stability conditions are valid.

Problem 2: Is $S(T, V, N)$ fundamental? (adapted from Callen)***

We call the functions $S(U, V, N)$ and $U(S, V, N)$ fundamental because from each of them encode complete thermodynamic information: from them we can derive all thermodynamic potentials as well as equation of states. In this problem we will demonstrate the if we express entropy as a function of T, V, N , the resulting function is no longer fundamental. Let us consider the fundamental function (2) again, this time shifted in energy by an arbitrary amount U_0 :

$$S(U, V, N) = AN \ln \left(\frac{(U - U_0)^{3/2} V}{N^{5/2}} \right). \quad (2)$$

Express the entropy as a function of T, V, N , and show that $S(T, V, N)$ is independent of U_0 . Therefore the information of U_0 is completely lost once we make variable transform from S, V, N to T, V, N —we can no longer extract all thermodynamic informations.

Lesson we should always try to express thermodynamic functions in terms of their natural variables.

Solution

We are given a shifted entropy function in the energy representation:

$$S(U, V, N) = AN \ln \left(\frac{(U - U_0)^{3/2} V}{N^{5/2}} \right)$$

We want to express this entropy as a function of the unnatural variables T, V, N . First, we compute the temperature by taking the partial derivative with respect to U :

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V, N} = \frac{3}{2} \frac{AN}{U - U_0} \implies U - U_0 = \frac{3}{2} ANT$$

Now we substitute the expression for $(U - U_0)$ back into the original entropy function to transition to the variables T, V, N :

$$\begin{aligned} S(T, V, N) &= AN \ln \left(\frac{\left(\frac{3}{2} ANT \right)^{3/2} V}{N^{5/2}} \right) \\ &= AN \ln \left(\left(\frac{3}{2} AT \right)^{3/2} \frac{V}{N} \right) \end{aligned}$$

As explicitly shown, the resulting function $S(T, V, N)$ depends only on A, T, V, N . The constant U_0 has been completely eliminated and intrinsically absent from this relation. Because U_0 is fundamentally lost, we cannot invert this transformation to recover the fundamental function $U(S, V, N)$; an arbitrary integration constant has disappeared. This verifies that $S(T, V, N)$ encapsulates incomplete thermodynamic information and is *not* a fundamental relation.

Problem 3: Energy Function for Polytropic Adiabats (adapted from Callen)★★

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Consider a single-component system in which adiabatic processes obey a polytropic relation of the form:

$$PV^k = \text{constant},$$

where $k > 0$ is a given constant. Show that the internal energy per mole can be written in the form

$$U(S, V, N) = \frac{1}{k-1}PV + Nf\left(\frac{PV^k}{N^k}\right),$$

where f is an arbitrary function of a single variable.

Solution

An adiabatic process in a simple, single-component thermodynamic system corresponds to paths where entropy S and particle number N are held constant. Therefore, the given polytropic relation implies that:

$$PV^k = c(S, N)$$

where $c(S, N)$ is a positive function dependent strictly on the constants of the adiabatic path. We can express the pressure as $P = c(S, N)V^{-k}$.

From the fundamental thermodynamic identity in the energy representation, $dU = TdS - PdV + \mu dN$, we know that:

$$P = -\left(\frac{\partial U}{\partial V}\right)_{S, N} \implies \left(\frac{\partial U}{\partial V}\right)_{S, N} = -c(S, N)V^{-k}$$

Integrating this expression with respect to V (holding S, N constant, and assuming $k > 0, k \neq 1$), we obtain the internal energy:

$$U(S, V, N) = \frac{c(S, N)}{k-1}V^{1-k} + g(S, N) \quad (3)$$

where $g(S, N)$ is a constant of integration with respect to volume. Recognizing that $c(S, N)V^{-k} = P$, we can substitute $c(S, N)V^{1-k} = PV$ back into the integral:

$$U(S, V, N) = \frac{1}{k-1}PV + g(S, N) \quad (4)$$

To determine the functional form of $g(S, N)$ and $c(S, N)$, we must use Euler's homogeneity requirements. Internal energy $U(S, V, N)$ is extensive and thus a homogeneous function of degree 1. Both variables P (intensive, degree 0) and V (extensive, degree 1) mean that the product PV has degree 1. For Eq. (9) to satisfy homogeneity, $g(S, N)$ must also be homogeneous of degree 1 in (S, N) . We can safely write it as $g(S, N) = N\psi(S/N)$.

Meanwhile, the polytropic constant $c(S, N) = PV^k$ must be a homogeneous function of degree $0 + k \times 1 = k$. Hence, it must take the form $c(S, N) = N^k\phi(S/N)$. This dictates that the ratio $c(S, N)/N^k$ is solely a function of S/N , implying $S/N = \phi^{-1}\left(\frac{c(S, N)}{N^k}\right)$.

Since ψ is an arbitrary function of S/N , we can compose it with ϕ^{-1} into a new arbitrary function f , yielding:

$$\psi(S/N) = f\left(\frac{c(S, N)}{N^k}\right) = f\left(\frac{PV^k}{N^k}\right)$$

Substituting this back into $g(S, N) = N\psi(S/N)$ leads to $g(S, N) = Nf\left(\frac{PV^k}{N^k}\right)$. Finally, replacing $g(S, N)$ in Eq. (9), we show exactly that:

$$U(S, V, N) = \frac{1}{k-1}PV + Nf\left(\frac{PV^k}{N^k}\right)$$

Problem 4: Partial derivatives***

Consider three variables related to each other via one equation: $X = X(Y, Z)$ or equivalently $Z = Z(X, Y)$, so that only two of them are independent. We can treat any of the three variables as a function of the other two. We can also write differential forms such as:

$$dZ = \left(\frac{\partial Z}{\partial X} \right)_Y dX + \left(\frac{\partial Z}{\partial Y} \right)_X dY \quad (5)$$

and other similar relations also follow. Using the differential form of various functions, prove the following relations:

$$\left(\frac{\partial X}{\partial Y} \right)_Z = 1 / \left(\frac{\partial Y}{\partial X} \right)_Z, \quad (6)$$

$$\left(\frac{\partial Z}{\partial Y} \right)_X = - \left(\frac{\partial X}{\partial Y} \right)_Z \left(\frac{\partial Z}{\partial X} \right)_Y \quad (7)$$

Solution

Let the given functions be $X(Y, Z)$ and $Y(X, Z)$. We can write their exact differentials as:

$$dX = \left(\frac{\partial X}{\partial Y} \right)_Z dY + \left(\frac{\partial X}{\partial Z} \right)_Y dZ \quad (8)$$

$$dY = \left(\frac{\partial Y}{\partial X} \right)_Z dX + \left(\frac{\partial Y}{\partial Z} \right)_X dZ \quad (9)$$

To relate the partial derivatives, we substitute Eq. (13) into Eq. (12):

$$\begin{aligned} dX &= \left(\frac{\partial X}{\partial Y} \right)_Z \left[\left(\frac{\partial Y}{\partial X} \right)_Z dX + \left(\frac{\partial Y}{\partial Z} \right)_X dZ \right] + \left(\frac{\partial X}{\partial Z} \right)_Y dZ \\ dX &= \left[\left(\frac{\partial X}{\partial Y} \right)_Z \left(\frac{\partial Y}{\partial X} \right)_Z \right] dX + \left[\left(\frac{\partial X}{\partial Y} \right)_Z \left(\frac{\partial Y}{\partial Z} \right)_X + \left(\frac{\partial X}{\partial Z} \right)_Y \right] dZ \end{aligned}$$

Because the variables X and Z can be chosen as mathematically independent parameters, their differentials dX and dZ are strictly linearly independent. By comparing the coefficients on the left-hand side and right-hand side, we establish two constraints:

1. From the coefficient of dX :

$$1 = \left(\frac{\partial X}{\partial Y} \right)_Z \left(\frac{\partial Y}{\partial X} \right)_Z \implies \left(\frac{\partial X}{\partial Y} \right)_Z = 1 / \left(\frac{\partial Y}{\partial X} \right)_Z$$

This establishes the standard reciprocal theorem.

2. From the coefficient of dZ :

$$0 = \left(\frac{\partial X}{\partial Y} \right)_Z \left(\frac{\partial Y}{\partial Z} \right)_X + \left(\frac{\partial X}{\partial Z} \right)_Y \quad (10)$$

By invoking the reciprocal theorem proved above, we can rewrite Eq. (14) using the inverse properties for variables sets:

$$\left(\frac{\partial Y}{\partial Z} \right)_X = \frac{1}{\left(\frac{\partial Z}{\partial Y} \right)_X} \quad \text{and} \quad \left(\frac{\partial X}{\partial Z} \right)_Y = \frac{1}{\left(\frac{\partial Z}{\partial X} \right)_Y}$$

Substituting these into Eq. (14):

$$\left(\frac{\partial X}{\partial Y}\right)_Z \frac{1}{\left(\frac{\partial Z}{\partial Y}\right)_X} + \frac{1}{\left(\frac{\partial Z}{\partial X}\right)_Y} = 0 \quad \Rightarrow \quad \frac{1}{\left(\frac{\partial Z}{\partial Y}\right)_X} = -\frac{1}{\left(\frac{\partial X}{\partial Y}\right)_Z \left(\frac{\partial Z}{\partial X}\right)_Y}$$

Inverting both sides finishes the cyclic chain rule proof:

$$\left(\frac{\partial Z}{\partial Y}\right)_X = -\left(\frac{\partial X}{\partial Y}\right)_Z \left(\frac{\partial Z}{\partial X}\right)_Y$$

Problem 5: Partial derivatives continued***

Suppose we have another variable W , which can also be understood as function of X, Y or as function of X, Z , etc. Prove:

$$\left(\frac{\partial W}{\partial Z}\right)_X = \left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial Z}\right)_X, \quad (11)$$

$$\left(\frac{\partial W}{\partial X}\right)_Z = \left(\frac{\partial W}{\partial X}\right)_Y + \left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial X}\right)_Z. \quad (12)$$

Solution

Suppose we treat W first as a function of the independent variables X and Y , namely $W(X, Y)$. The exact differential reads:

$$dW = \left(\frac{\partial W}{\partial X}\right)_Y dX + \left(\frac{\partial W}{\partial Y}\right)_X dY \quad (13)$$

We can also treat Y as a function of independent variables X and Z , taking the exact differential:

$$dY = \left(\frac{\partial Y}{\partial X}\right)_Z dX + \left(\frac{\partial Y}{\partial Z}\right)_X dZ \quad (14)$$

To transform the dependencies of W towards coordinates X and Z , we substitute Eq. (18) into Eq. (17):

$$\begin{aligned} dW &= \left(\frac{\partial W}{\partial X}\right)_Y dX + \left(\frac{\partial W}{\partial Y}\right)_X \left[\left(\frac{\partial Y}{\partial X}\right)_Z dX + \left(\frac{\partial Y}{\partial Z}\right)_X dZ \right] \\ dW &= \left[\left(\frac{\partial W}{\partial X}\right)_Y + \left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial X}\right)_Z \right] dX + \left[\left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial Z}\right)_X \right] dZ \end{aligned}$$

If we independently write out the exact differential form for W mapped natively as a function of X and Z , $W(X, Z)$, we have:

$$dW = \left(\frac{\partial W}{\partial X}\right)_Z dX + \left(\frac{\partial W}{\partial Z}\right)_X dZ$$

Since X and Z act as linearly independent variables, the coefficients standing in front of dX and dZ for the two equivalent dW statements must be algebraically strictly equivalent. Equating the coefficient of dZ , we obtain the first relation directly:

$$\left(\frac{\partial W}{\partial Z}\right)_X = \left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial Z}\right)_X$$

Equating the coefficient of dX , we obtain the second relation naturally:

$$\left(\frac{\partial W}{\partial X}\right)_Z = \left(\frac{\partial W}{\partial X}\right)_Y + \left(\frac{\partial W}{\partial Y}\right)_X \left(\frac{\partial Y}{\partial X}\right)_Z$$

Problem 6: Homogeneity and the flat direction***

The thermodynamic entropy $S(U, V, N)$ is homogeneous of degree one:

$$S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N). \quad (15)$$

1. Taking two derivatives of Eq. (15) with respect to λ , show that the following quadratic form vanishes identically:

$$\begin{pmatrix} U & V & N \end{pmatrix} \begin{pmatrix} S_{UU} & S_{UV} & S_{UN} \\ S_{UV} & S_{VV} & S_{VN} \\ S_{UN} & S_{VN} & S_{NN} \end{pmatrix} \begin{pmatrix} U \\ V \\ N \end{pmatrix} = 0. \quad (16)$$

Recall that $S(U, V, N)$ is concave. Show that Eq. (16) means that the Hessian of $S(U, V, N)$ has a zero eigenvalue, with eigenvector proportional to (U, V, N) .

2. Divide an isolated thermodynamic system into two identical subsystems, and allow them to exchange energy, volume, and particles. Interpret the physical meaning of the zero eigenvalues of the Hessian shown above.
3. Now express the entropy as

$$S(U, V, N) = N s(u, v) = N s(U/N, V/N),$$

and explicitly show that the Hessian matrix of $S(U, V, N)$ is indeed singular.

Solution

1. Because entropy $S(U, V, N)$ is extensive, it is a homogeneous function of degree one, following Euler's theorem:

$$S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N)$$

Taking the first derivative of both sides with respect to λ yields Euler's homogeneous identity:

$$US_U(\lambda U, \lambda V, \lambda N) + VS_V(\lambda U, \lambda V, \lambda N) + NS_N(\lambda U, \lambda V, \lambda N) = S(U, V, N)$$

Taking a second derivative with respect to λ effectively applies the chain rule a second time over the gradient:

$$U[US_{UU}(\dots) + VS_{UV}(\dots) + NS_{UN}(\dots)] + V[US_{VU}(\dots) + VS_{VV}(\dots) + NS_{VN}(\dots)] \\ + N[US_{NU}(\dots) + VS_{NV}(\dots) + NS_{NN}(\dots)] = 0$$

Evaluating this relation at physically meaningful scale $\lambda = 1$ consolidates to:

$$U^2 S_{UU} + V^2 S_{VV} + N^2 S_{NN} + 2UV S_{UV} + 2UN S_{UN} + 2VN S_{VN} = 0$$

which can be written elegantly into quadratic matrix form involving the Hessian matrix (H):

$$\begin{pmatrix} U & V & N \end{pmatrix} \begin{pmatrix} S_{UU} & S_{UV} & S_{UN} \\ S_{UV} & S_{VV} & S_{VN} \\ S_{UN} & S_{VN} & S_{NN} \end{pmatrix} \begin{pmatrix} U \\ V \\ N \end{pmatrix} = 0$$

Because thermodynamic stability obligates the entropy to be purely concave, H strictly forms a *negative semi-definite* matrix ($x^T H x \leq 0$). The only mathematical means for a negative semi-definite quadratic form to equal identically zero is if the constituent vector resides within its null-space, $Hx = 0$. Consequently, the vector $(U, V, N)^T$ stands as an eigenvector corresponding to an eigenvalue of zero.

2. Physical Interpretation:

If an isolated macroscopic system is bifurcated into identically featured subsystems permitted to freely exchange extensive resources (U, V, N) , the local change in total entropy for small fluctuations relies upon the Hessian. The specific zero eigenvalue corresponding strictly to the eigenvector (U, V, N) means that proportional perturbations ($\delta U \propto U, \delta V \propto V, \delta N \propto N$) strictly yield zero alteration into total overarching entropy ($\Delta S = 0$). Physically, moving along this "flat direction" corresponds to uniformly altering the macroscopic footprint of boundaries between identical subsystems, shrinking one side while equivalently enlarging the other. Their intensive densities remain statically identical, representing pure translational phase invariance lacking entropy penalties natively tied to scale equivalence.

3. Utilizing density representation: $u = U/N$ and $v = V/N$ alongside $S(U, V, N) = Ns(u, v)$. The scaled primary derivatives of S trace to:

$$S_U = s_u, \quad S_V = s_v, \quad S_N = s - us_u - vs_v$$

Next, evaluating elements mapping to the complete 3×3 Hessian matrix H :

$$\begin{aligned} S_{UU} &= \frac{1}{N} s_{uu}, & S_{UV} &= \frac{1}{N} s_{uv}, & S_{VV} &= \frac{1}{N} s_{vv} \\ S_{UN} &= \frac{\partial S_U}{\partial N} = -\frac{1}{N} (us_{uu} + vs_{uv}), & S_{VN} &= \frac{\partial S_V}{\partial N} = -\frac{1}{N} (us_{vu} + vs_{vv}) \\ S_{NN} &= \frac{\partial S_N}{\partial N} = -\frac{1}{N} (us_u + vs_v) - uS_{UN} + s_u \frac{u}{N} - vS_{VN} + s_v \frac{v}{N} = \frac{1}{N} (u^2 s_{uu} + 2uv s_{uv} + v^2 s_{vv}) \end{aligned}$$

The fully populated Hessian writes down as follows:

$$H = \frac{1}{N} \begin{pmatrix} s_{uu} & s_{uv} & -(us_{uu} + vs_{uv}) \\ s_{uv} & s_{vv} & -(us_{uv} + vs_{vv}) \\ -(us_{uu} + vs_{uv}) & -(us_{uv} + vs_{vv}) & u^2 s_{uu} + 2uv s_{uv} + v^2 s_{vv} \end{pmatrix}$$

Letting C_1, C_2, C_3 identify the columns from left to right, we can test linear dependency utilizing combinations spanning $C_3 = -uC_1 - vC_2$:

- Row 1: $-u(s_{uu}) - v(s_{uv}) = -(us_{uu} + vs_{uv})$ (matches C_3)

- Row 2: $-u(s_{uv}) - v(s_{vv}) = -(us_{uv} + vs_{vv})$ (matches C_3)
- Row 3: $-u[-(us_{uu} + vs_{uv})] - v[-(us_{uv} + vs_{vv})] = u^2s_{uu} + 2uv s_{uv} + v^2s_{vv}$ (matches C_3)

Because the final column is merely a strict linear combination of the first two corresponding to density multipliers (u, v) , the rank of the matrix is at most 2. Hence the determinant equals strictly zero, and we've verified inherently that the Hessian matrix of $S(U, V, N)$ is singular.